

Diagrammatic equation for the operator $\hat{S}^{0,1}$:

The left side shows a vertical line with two horizontal lines crossing it, each marked with an 'X' at the intersection. This is equal to a square box containing two arcs connecting the top and bottom horizontal lines. This is equal to the operator $\hat{S}^{0,1}$ defined by the sum over $j, k = 0$ to $d-1$ of $|j\rangle\langle k| \otimes |k\rangle\langle j|$.

$$= \text{[Diagram: Square box with two arcs]} \iff \hat{S}^{0,1} = \sum_{j,k=0}^{d-1} |j\rangle\langle k| \otimes |k\rangle\langle j|$$